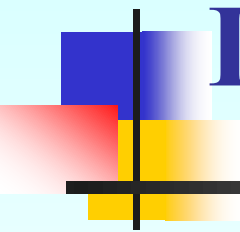
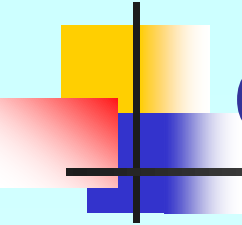


Lecture 1

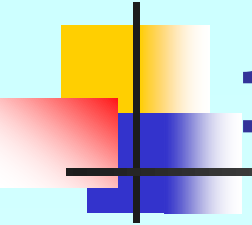


Introduction to Spatial Analysis



Outline

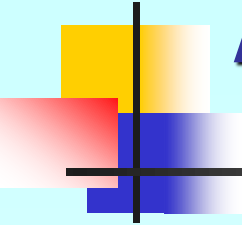
- 1. What is "Spatial Analysis"?**
- 2. Some concepts in SA**
- 3. Why SA Is Important?**
- 4. The Frame of Spatial Analysis**
- 5. How Spatial Analysis Work?**



1. What is “Spatial Analysis”

- **Introduction: a case**

- **Definitions**



An Early Example:

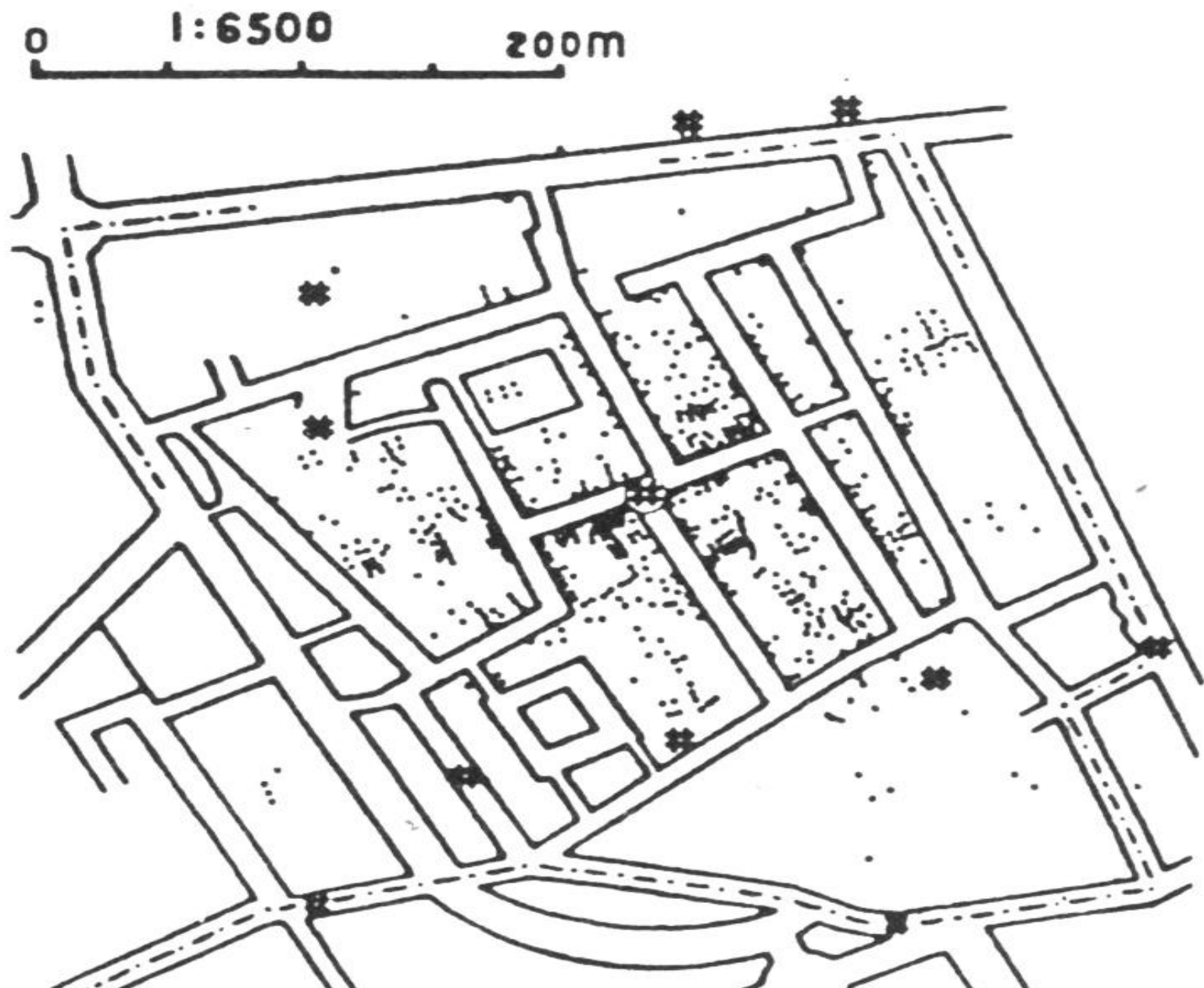
John Snow's Cholera Investigation

- **1854年8月到9月**英国伦敦霍乱流行时，当局始终找不到发病的原因，后来医生约翰·斯诺 (John Snow) 参与调查。
- 他在绘有霍乱流行地区所有道路、房屋、饮用水机井等内容的1：6500比例尺地图上，标出了每个霍乱病死者的**居住位置**，得到了**霍乱病死者居住分布图**。

John Snow (1813.3.15 – 1858.6.16)

A British physician 内科医生 and a leader in the adoption of anaesthesia 麻醉 and medical hygiene 医学保健. He is considered to be one of the fathers of epidemiology 流行病学, because of his work in tracing the source of a cholera 霍乱 outbreak in Soho, England, in 1854.

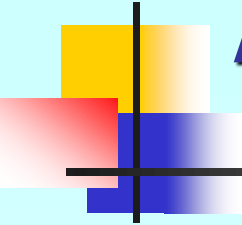




霍乱病死者居住分布图(John Snow, 1854)



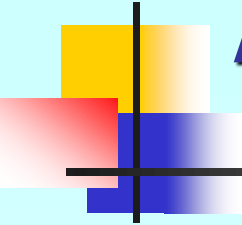
霍乱死亡病例与水源的关系



An Early Example:

John Snow's Cholera Investigation

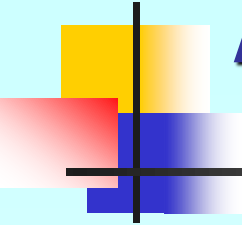
- 斯诺博士分析了这张分布图，马上想到霍乱病源之所在
 - - 死者住家都集中于饮用 **“布洛多斯托” 井水** 的地方及周围。



An Early Example:

John Snow's Cholera Investigation

- 根据斯诺博士的分析和请示，当局于9月8日摘下了这个水井的泵，禁止使用该水泵吸水，从这天以后，新的霍乱病患者就再也没有出现了。
- 在这个例子中，患者的**居住地与饮用水井之间的空间位置关系**提示了霍乱病的发病根源。



An Early Example:

John Snow's Cholera Investigation

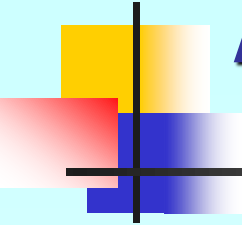
- 1) 收集存活者与死亡的生活情况
- 2) 霍乱病是消化道流行病，与饮食有关
- 3) 居民在水泵取水喝
- 4) 假设水泵向周围的居民提供了不洁的水
- 5) 病人居住地的分布围绕着其中一个水泵
- 6) 该水泵受到了污染
- 7) 关闭该水井

An Early Example:

John Snow's Cholera Investigation

There is a plaque commemorating Snow and his 1854 study in the place of the water pump on Broadwick Street with a water pump with its handle removed, near what is now "The John Snow" public house.





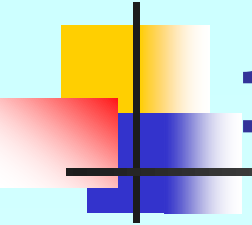
An Early Example:

John Snow's Cholera Investigation

Review: In today's terminology, this was a simple **dot map** and made possible by Snow's access to individual case addresses. Viewed on its own, this map showed some evidence of **spatial clustering** of cases of the disease, but following a hunch, Snow **overlaid** a second map of water pumps in the area and was able to show that many cases were **concentrated around** a single source of infected water.

The discovery of a link between cholera and water supply led to a movement in the United Kingdom toward public water supply in the second half of the nineteenth century.

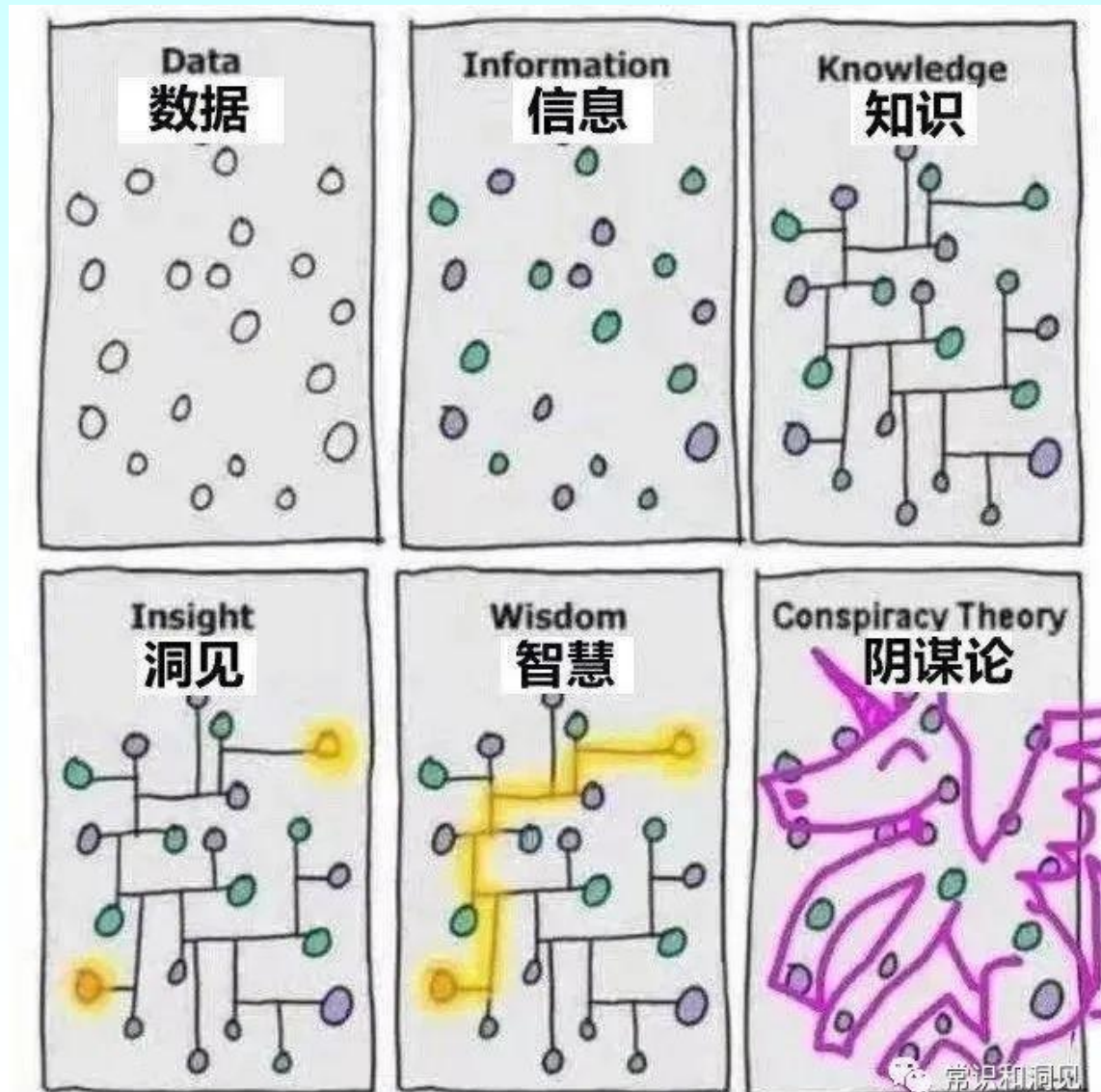
--David O'sullivan and David J. Unwin (Geographic Information Analysis, 2003)

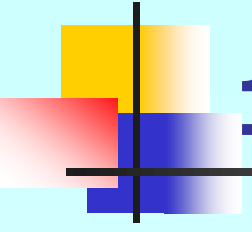


1. What is “Spatial Analysis”

**Brain storm about the
definition of SA**

Analysis: from data to knowledge

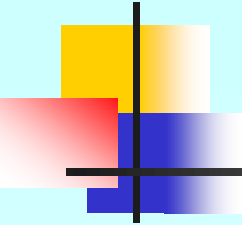




1. What is “Spatial Analysis” Definition 1

- 空间分析：是基于地理对象的位置和形态特征的空间数据分析技术，其目的在于提取和传输空间**信息/知识**。

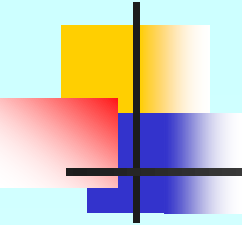
——郭仁忠 《空间分析》，2001.



1. What is “Spatial Analysis” Definition 2

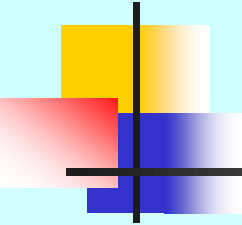
指分析具有空间坐标或相对位置的数据和过程的理论和方法，与其他数据分析方法比较，八个特性（王劲峰，2006）：

1. 空间数据中普遍存在的空间相关性使其与经典统计学经常要求的**样本独立性**前提相悖，直接造成使用经典统计学分析空间数据得到的结论是有偏的和非最优的。
2. 空间数据一般具有不可重复性，观察到的数据只是空间过程的一次实现，这与物理、化学、生物等可实验科学不同，因此，必须在一定的**假设条件**下，空间数据才是可统计的（Haining, 2003; Isaaks, 1989）。



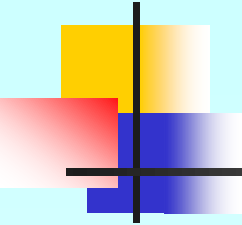
1. What is “Spatial Analysis” Definition 2

3. 考虑数据的空间维使其信息量更加丰富和细致，空间分析擅长于描述和揭示这些数据由所蕴涵的独特的空间信息、**关系、格局和过程**。
4. 空间分析技术是完成某些任务的唯一手段，例如，地形分析、地图使用和基于位置的服务（**LBS**），虽然这些方法古已有之，但是在现代地理信息系统的支持下，可以**定量化、自动化和智能化**。
5. 一些表面现象的背后以空间格局为其发生**机理**。



1. What is “Spatial Analysis” Definition 2

6. 空间是事物存在和人文与自然交互的界面，因此**空间成为多源信息的索引项**及人地关系界面。
7. 空间维增加了运筹和干预的搜索空间，有更多选择，使系统**调控**更加优化。
8. 空间维与其他维的相互作用与转化：当前的空间格局是其随后演化的边界条件，过程演化进程的空间不平衡可以被用来以空间反演**时间演化**，以时间演化树推断空间状态。



1. What is “Spatial Analysis” Definition X

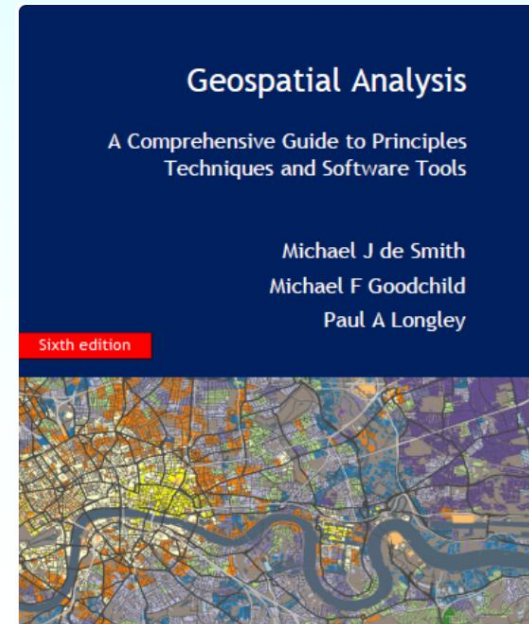
Spatial analysis is a process in which you **model problems geographically**, derive results by computer processing, and then explore and examine those results. The true power of GIS lies in the ability to perform analysis.

<https://www.esri.com/arcgis-blog/products/product/analytics/how-to-perform-spatial-analysis>

Chapter 2

Conceptual Frameworks for Spatial Analysis

Geospatial analysis provides **a distinct perspective** on the world, **a unique lens** through which to examine events, patterns, and processes that operate on or near the surface of our planet. It **makes sense**, then, to introduce the main elements of this perspective, the conceptual framework that provides the background to spatial analysis, as a preliminary to the main body of this Guide’s material.



1. What is “Spatial Analysis” Definition X



Harvard University
Graduate School of Design

Courses

Spatial Analysis

Planning decisions are often idealized as being "evidence-based" or "data-driven." Spatial data often comprise the data and evidence that support such these decisions. In this course, you will learn how to create spatial datasets, both by assembling them from existing sources and by collecting data in the field. You will also learn data visualization techniques to identify spatial relationships within and between spatial datasets. You will learn to identify the arguments that mapmakers make through their maps and to use spatial data visualization to make your own arguments. Student in this course are expected to have completed the preterm workshop for MUP students and software tutorials on ArcGIS, R, and RStudio.

Menu Search GSD Now

Course Number

VIS-2128

Hours/Location

Tue, Thu 9:00 AM – 10:15 AM @ Gund 516

Instruction Mode

In Person

Fall 2022

Lecture

2 Credits

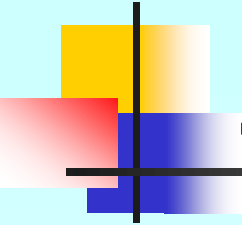
↗ [Course Website](#)

Instructor

[Carole Voulgaris](#)

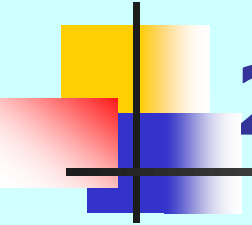
Department

[Department of Urban Planning and Design](#)



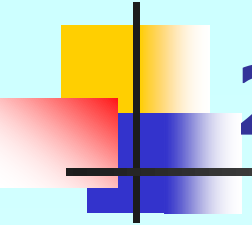
2. Some concepts in SA

- Spatial Data
- Spatial Location
- Spatial Distribution
- Spatial Morphology
- Spatial Relationship



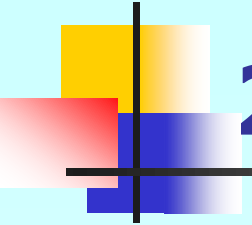
2. Some concepts in SA

- **数据的空间分析：** 直接从空间物体的空间位置、联系等方面去研究空间事物， 以期对空间事物作出定量的描述。
 - 图论、拓扑学
 - 网络分析， 空间聚类， 热力图heat map
- **空间数据的分析：** 属性数据的空间分布特征分析
 - 计量地理、空间统计， 依托于地理空间进行解释
 - 热点分析 hot spots



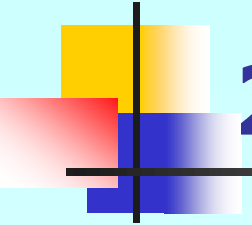
2. Some concepts in SA

- The **domain of geospatial analysis** is the surface of the Earth, extending upwards in the analysis of topography and the atmosphere, and downwards in the analysis of groundwater and geology.
- **In scale** it extends from the most local, when archaeologists record the locations of pieces of pottery to the nearest centimeter or property boundaries are surveyed to the nearest millimeter, to the global, in the analysis of sea surface temperatures or global warming.



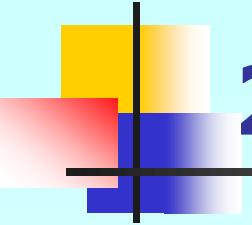
2. Some concepts in SA

- In time it extends backwards from the present into the analysis of historical population migrations, the discovery of patterns in archaeological sites, or the detailed mapping of the movement of continents, and into the future in attempts to predict the tracks of hurricanes, the melting of the Greenland ice-cap, or the likely growth of urban areas.



2. Some concepts in SA

- **Basic Primitives:** the basic components of this view of the world — the classes of things.
- **Spatial Relationships:** some of the structures that are built with these basic components and the relationships between them that interest geographers and others.



2. Some concepts in SA

Basic Primitives

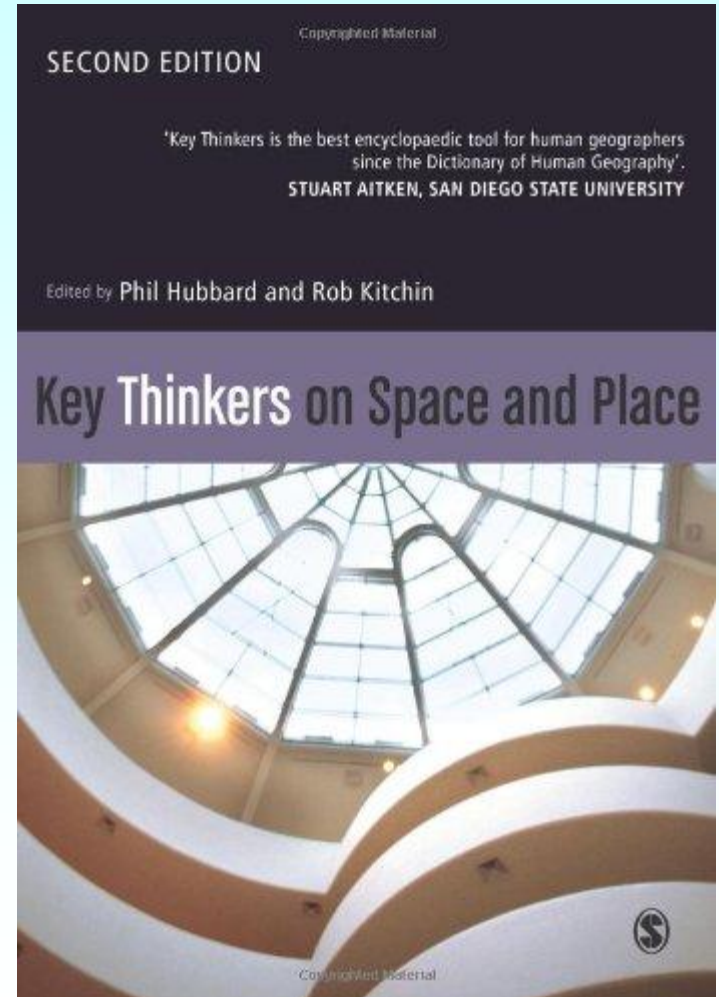
- Space and Places (微博应用)
- Attributes
- Objects
- Maps (映射)
- Multiple properties of places
- Fields (场模型)
- Networks
- Density estimation
- Detail, resolution, and scale
- Topology

2. Some concepts in SA

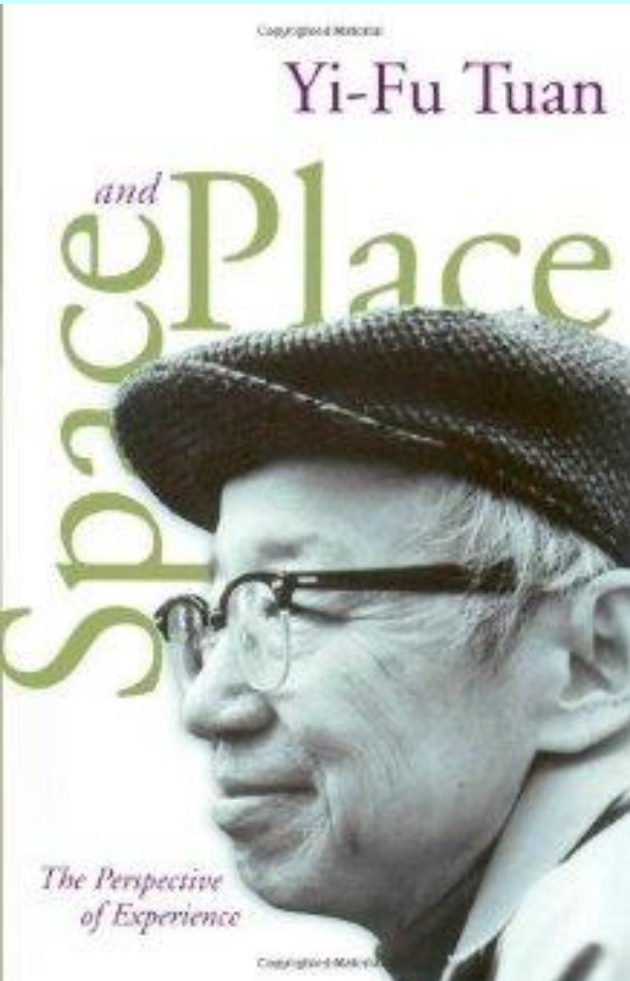
■ Space and Place

石崧, 宁越敏. 人文地理学“空间”
内涵的演进. 地理科学, 2005,
25(3): 340-345.

■ Spatial Cognition



Space and Place: The Perspective of Experience

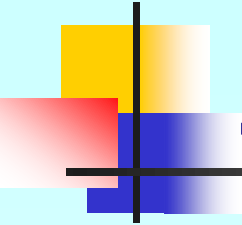


Geography

On the 25th anniversary of its publication, a new edition of this foundational work on human geography.

In the twenty years since its original publication, *Space and Place* has not only established the discipline of human geography, but it has proven influential in such diverse fields as theatre, literature, anthropology, psychology, and theology. Eminent geographer Yi-Fu Tuan considers the ways in which people feel and think about space, how they form attachments to home, neighborhood, and nation, and how feelings about space and place are affected by the sense of time. He suggests that place is security and space is freedom: we are attached to the one and long for the other. Whether he is considering sacred versus "biased" space, mythical space and place, time in experiential space, or cultural attachments to space, Tuan's analysis is thoughtful and insightful throughout.

Until retiring in 1998, Yi-Fu Tuan was a professor of geography at the University of Wisconsin-Madison. He is ranked among the country's most distinguished cultural geographers and has earned numerous honors, among them a Guggenheim Fellowship, the Bracken Award for landscape architecture, and an award for meritorious contribution to geography from the Association of American Geographers. He was recently named the Lauréat d'Honneur 2000 of the International Geographers Union. He is the author of many essays and books, including *Escapism* (1998) and *Cosmos and Hearth* (Minnesota, 1999).

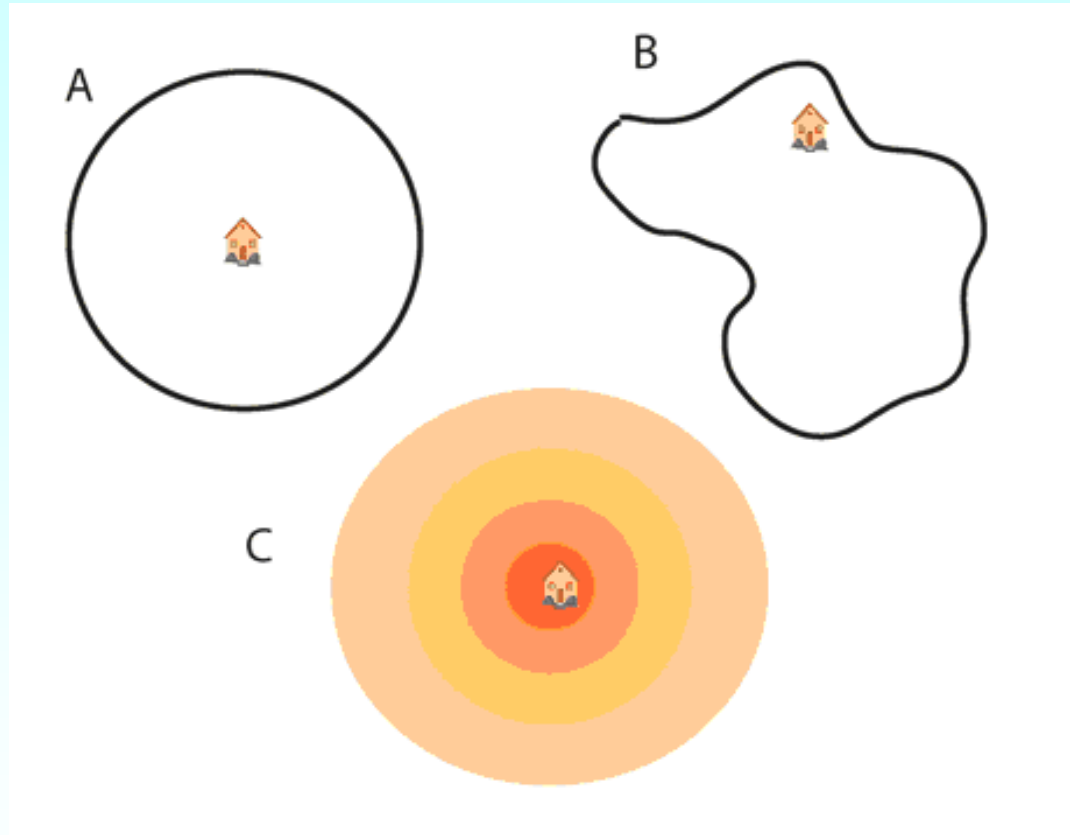


2. Some concepts in SA

Spatial Relationships

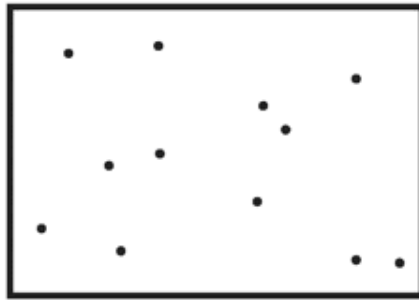
- Co-location
- Distance, direction and spatial weights matrices
- Multidimensional scaling
- Spatial context
- Neighborhood
- Spatial heterogeneity
- Spatial dependence
- Spatial sampling
- Spatial interpolation
- Smoothing and sharpening
- First- and second-order processes

2. Some concepts in SA



Three alternative ways of defining neighborhood, using simple GIS functions

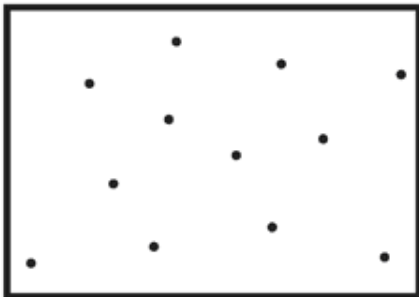
2. Some concepts in SA



A



B

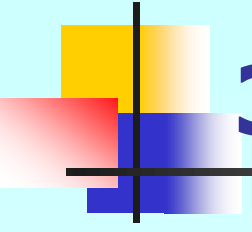


C



D

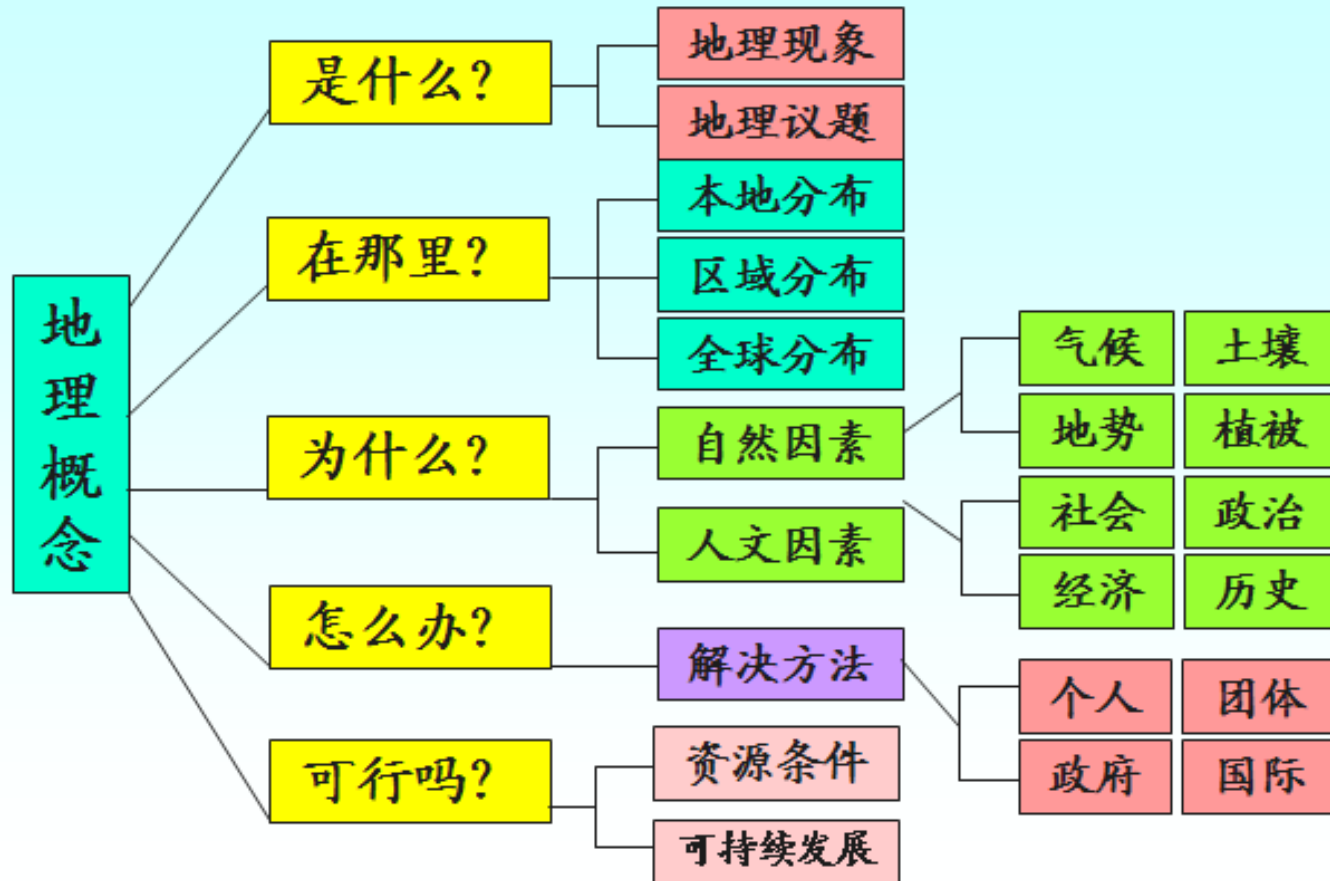
Four distinct patterns of twelve points in a study area
(A) random, (B) clustered, (C) dispersed, (D) clustered



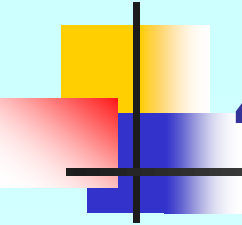
3. Why SA Is Important?

- What is the difference between GIS and MIS?
- What is the essential feature of GIS?
- What is Intelligent GIS?
- **Big Data**

4. How Spatial Analysis Work?



用于对地理概念的理解的概念图



4. How Spatial Analysis Work?

理论基础

- 地理学基础

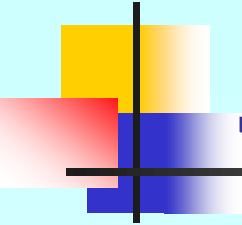
- 自然地理学
- 人文地理学：空间综合社会科学

- 数学基础

- 经典统计学
- 图论
- 运筹学

- 计算机科学基础

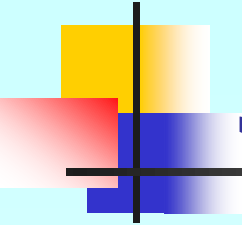
- 算法基础



5. The Frame of Spatial Analysis

一、基本理论

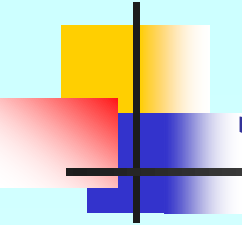
1. 空间分析的基础——空间量算、空间分布、空间形态描述
2. 空间分析的基础——空间关系与空间自相关
3. 空间分析的功能体系
4. 空间分析的算法基础
5. 空间分析应用模式



5. The Frame of Spatial Analysis

二、功能体系与软件工具

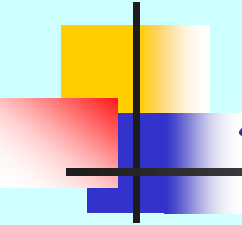
- 空间形态度量
- 空间统计
- 空间插值
- 景观格局分析
- 叠置分析、缓冲区分析、网络分析、数字高程模型分析（三维分析）
- 探索性空间数据分析
- 地图代数
- 区位—配置模型
- 决策树与决策支持技术
- 空间数据挖掘：离群点发现、模式发现
- 空间推理、时空推理、案例推理
- 人工生命与元胞自动机、人工神经网络、地理过程模拟
- 地理计算
- 机器学习
- 地理大数据、空间大数据



5. The Frame of Spatial Analysis

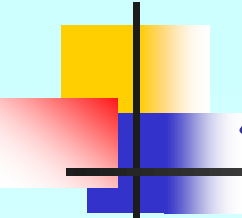
三、应用领域

1. 空间统计
2. 空间规划与设计
3. 工程计算
4. 景观指数及其应用：生态环境
5. 空间数据挖掘
6. 空间分析的社会科学应用：空间经济学、人文科学
7. 空间综合分析



总结与讨论1

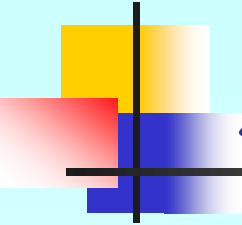
- Please enumerate all methods of spatial analysis you know and present taxonomy of these methods.



总结与讨论1

空间分析的内涵与外延？

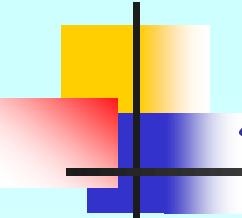
- ① 定义
- ② 功能框架
- ③ 有关软件
- ④ 典型案例



总结与讨论2

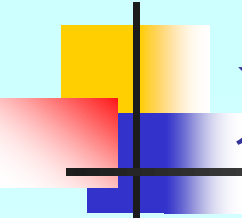
空间分析的PPDAC模式？

参见PPT: Methodological Context-- PPDAC:
Spatial Analysis as a Process 20220829.pdf



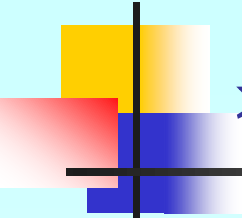
总结与讨论2

- ① 空间分析理论基础：模型、假设、定理
- ② 问题分析方法
- ③ 解决模型
- ④ 计算方法
- ⑤ 软件与算法问题
- ⑥ 结果分析与检验



延伸阅读

- 空间分析与GEOAI是什么关系？
- 武汉大学2025年新增的本科专业“时空信息工程”，其培养目标是“掌握空天地海一体化感知、时空认知、时空数据决策、时空智能体研发能力，服务数字中国、智慧城市、自动驾驶、低空经济、国防军事等国家战略”，据此，你认为要开设哪些空间分析相关课程？与地理信息科学专业有何区别？



参考书

- 空间分析（王劲峰等，2006）
- ARCGIS地理信息系统空间分析实验教程（汤国安等，2006）
- *Geographical Information Systems, Volume 1, Principles and Technical Issues*(2ED), Paul A. Longley, et al., 2004.
- GIS空间分析原理与方法（刘湘南等，2005）
- 空间分析(2ED)（郭仁忠，2002）
- GIS环境下的空间分析和地学视觉化（黄波等，2002.）